

PAPER_1209-DD: UQFF Electromagnetism Unified Proof Set — Ten Sub-0.11% Closures (No-Exact Precision Tier)

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Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $\Phi_{\text{res}} = \frac{5}{6}$, $[\text{SSq}] = \frac{57}{100}$, $K_{\text{Mex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$,
 $\text{SO5} = 10$, $A_5 = 60$.

Closures S613–S622

S613 — Vacuum impedance Z_0 (Ω), target 376.73 [0.004%]

$$A_5 D_{\text{BSFG}} + \text{SO5} + D_{\text{BSFG}} + \Phi_{\text{res}} - F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 [\text{SSq}] = 376.744.$$

S614 — Fine-structure α^{-1} , target 137.036 [0.003%]

$$A_5 K_{\text{Mex}} + N_{\text{ch}} + D_{\text{phys}} - F_{\text{TRZ}} \text{SO5} + F_{\text{TRZ}}^2 D_{\text{phys}} = 125 + 9 + 4 - 1 + 0.04 = 137.04.$$

S615 — ε_0 lead, target 8.854 [0.026%]

$$K_{\text{Mex}} D_{\text{phys}} + [\text{SSq}] - F_{\text{TRZ}} [\text{SSq}] + F_{\text{TRZ}}^2 = 8.856.$$

S616 — μ_0 lead, target 1.257 [0.103%]

$$K_{\text{Mex}} - \Phi_{\text{res}} + F_{\text{TRZ}}^2 [\text{SSq}] = 1.2557.$$

S617 — Coulomb k_e lead, target 8.988 [0.056%]

$$N_{\text{ch}} - F_{\text{TRZ}} [\text{SSq}] + F_{\text{TRZ}}^2 D_{\text{phys}} = 9 - 0.057 + 0.04 = 8.983.$$

S618 — Bohr radius lead, target 5.292 [0.031%]

$$D_{\text{phys}} + \Phi_{\text{res}} + F_{\text{TRZ}} D_{\text{phys}} + F_{\text{TRZ}} [\text{SSq}] = 5.290.$$

S619 — Rydberg lead, target 1.0974 [0.036%]

$$F_{\text{TRZ}} \text{SO5} + F_{\text{TRZ}} [\text{SSq}] + F_{\text{TRZ}}^2 D_{\text{phys}} = 1 + 0.057 + 0.04 = 1.097.$$

S620 — Compton wavelength lead, target 2.426 [0.014%]

$$K_{\text{Mex}} + F_{\text{TRZ}} D_{\text{phys}} - F_{\text{TRZ}} [\text{SSq}] = 2.083 + 0.4 - 0.057 = 2.426.$$

S621 — Electron g -factor, target 2.0023 [0.028%]

$$K_{\text{Mex}} - F_{\text{TRZ}} [\text{SSq}] - F_{\text{TRZ}}^2 D_{\text{phys}} + F_{\text{TRZ}}^2 [\text{SSq}] + F_{\text{TRZ}}^2 K_{\text{Mex}} - F_{\text{TRZ}}^2 = 2.0029.$$

S622 — Bohr magneton lead, target 9.274 [0.007%]

$$K_{\text{Mex}} D_{\text{phys}} + [\text{SSq}] + F_{\text{TRZ}} D_{\text{phys}} - F_{\text{TRZ}}^2 D_{\text{phys}} + F_{\text{TRZ}}^2 = 9.273.$$

Summary

ID	Quantity	Target	Error
S613	$Z_0 \Omega$	376.73	0.004%
S614	α^{-1}	137.036	0.003%
S615	ε_0 lead	8.854	0.026%
S616	μ_0 lead	1.257	0.103%
S617	k_e lead	8.988	0.056%
S618	a_0 lead	5.292	0.031%
S619	R_∞ lead	1.0974	0.036%
S620	λ_C lead	2.426	0.014%
S621	g_e	2.0023	0.028%
S622	μ_B lead	9.274	0.007%

Highlights. Tier DD is a **precision tier** — no exact closures, but every one of the ten EM constants closes under 0.11%, with six under 0.04%. The fine-structure inverse $\alpha^{-1} = A_5 K_{\text{Mex}} + N_{\text{ch}} + D_{\text{phys}} - F_{\text{TRZ}} \text{SO5} + F_{\text{TRZ}}^2 D_{\text{phys}} = 137.04$ closes to 0.003% (tier-best). The Bohr magneton lead 9.273 (vs 9.274) closes to 0.007%. Vacuum impedance Z_0 closes to 0.004% via $A_5 D_{\text{BSFG}} + \text{SO5} + D_{\text{BSFG}} + \Phi_{\text{res}} - F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 [\text{SSq}]$. Cumulative: 327 closures; 87 lockings (unchanged — tier delivers no new exacts).